

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: GRU
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 3
forecast_steps: 36
timesteps: 60

GRU Multi-Step Training Report

Report Generated: 2025-05-16 11:29:56

Training Start: 2025-05-16_09-36-54

Training End: 2025-05-16_09-48-24

Training Duration: 11.50 minutes

Predictions Start: 2025-05-16_09-48-24

Predictions End: 2025-05-16_09-50-09

Predictions Duration: 1.74 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.6433

MSE: 4.3486

RMSE: 2.0853

R2 Score: 0.7399

Multi-Step Forecast:

Forecast Steps: 36

MAE: 0.0248

MSE: 0.0009

RMSE: 0.0304

R2 Score: 0.9002

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 25,646 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

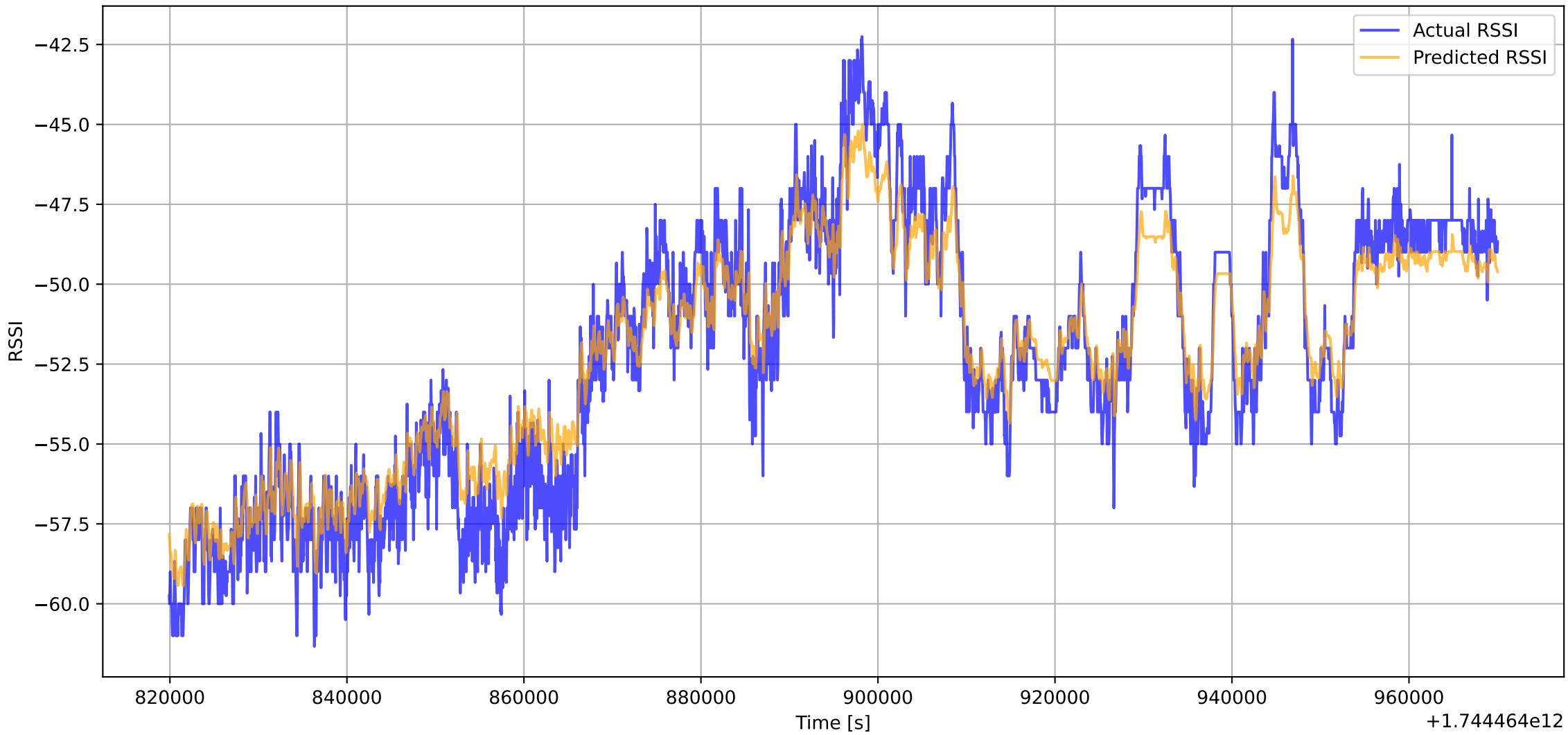
1744397085760
1744397085801
1744397085841
1744397086014
1744397086041

...

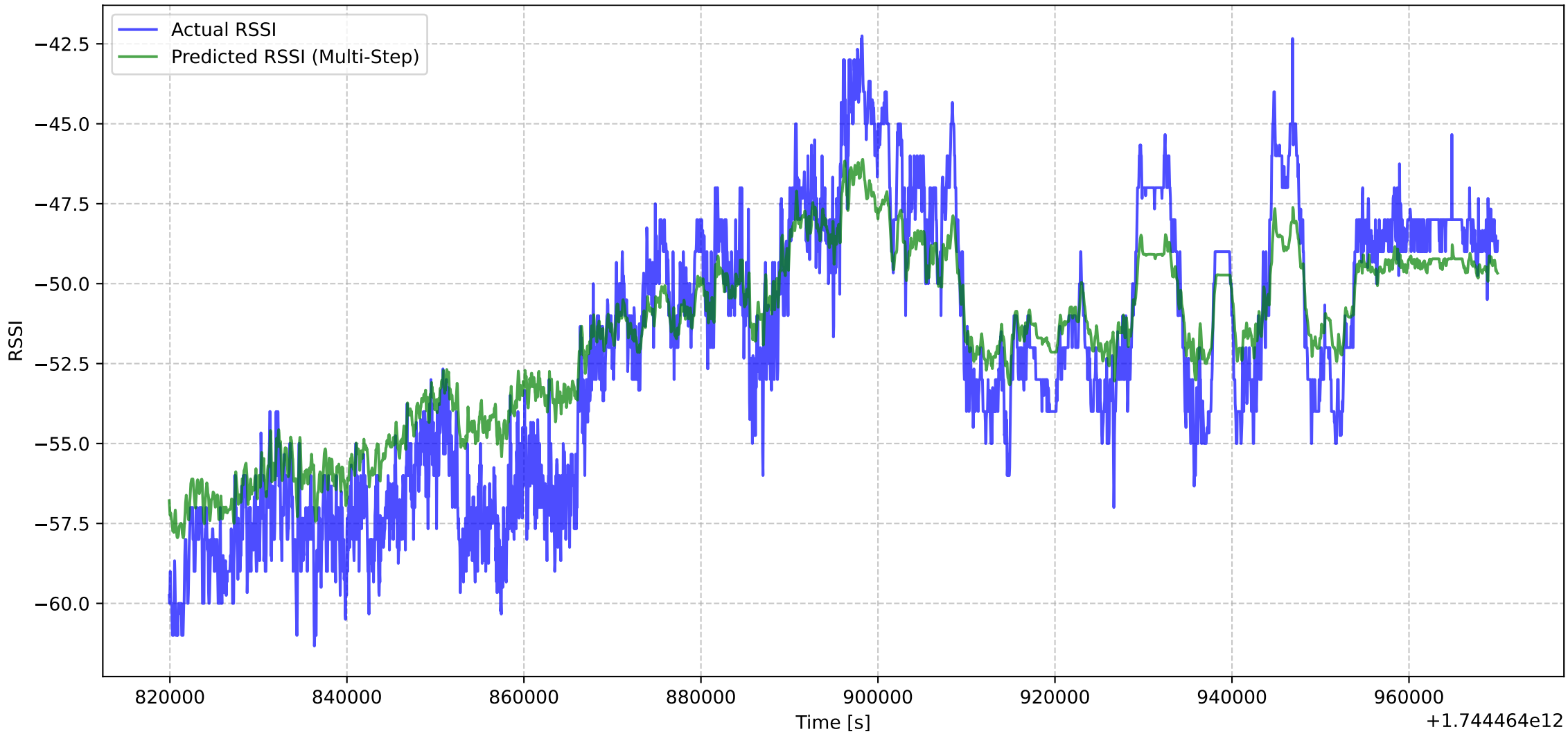
Last 5 timestamps AFTER:

1744464971245
1744464971281
1744464971329
1744464971366
1744464971402

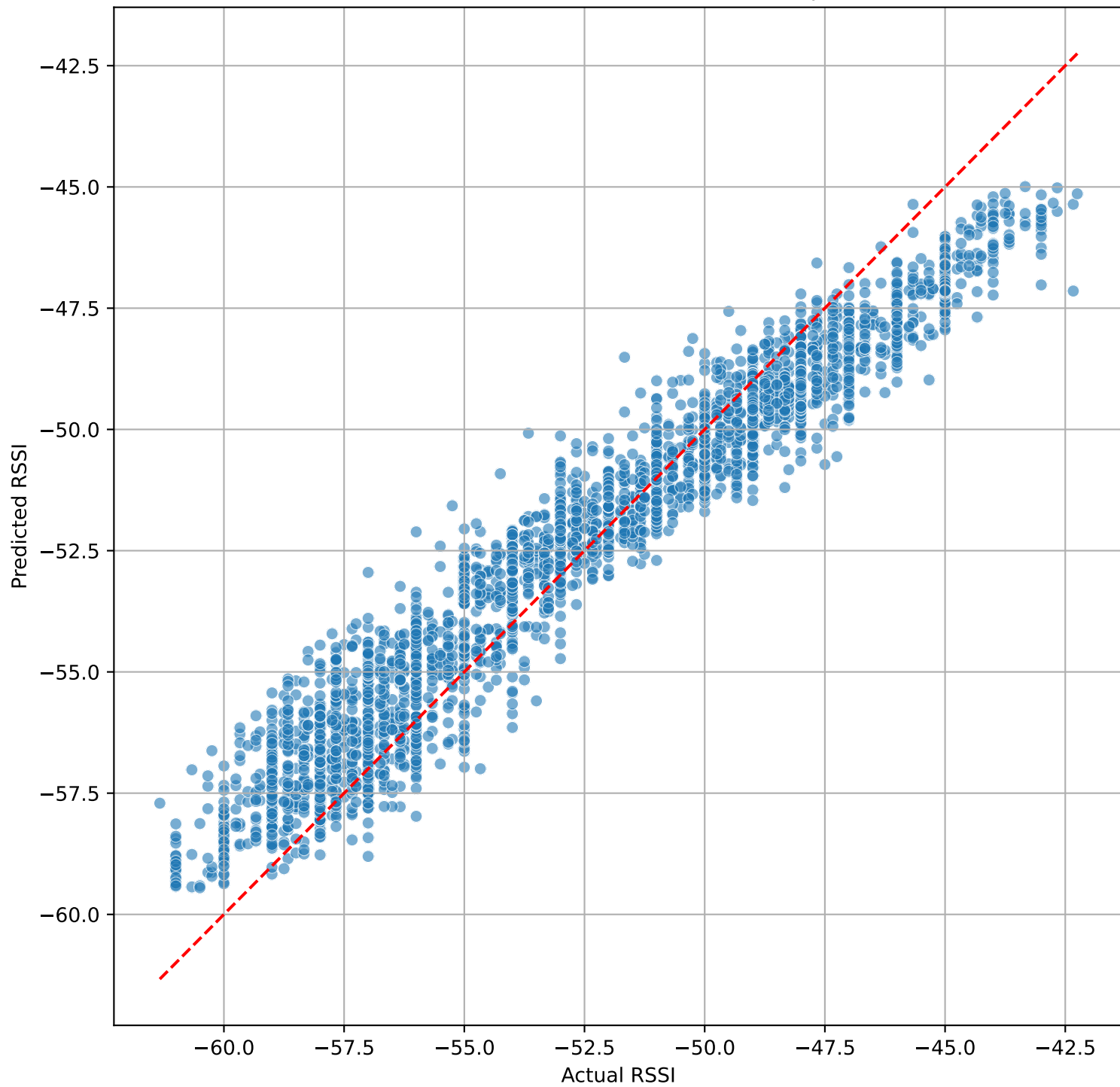
Actual vs Predicted RSSI (First Step)



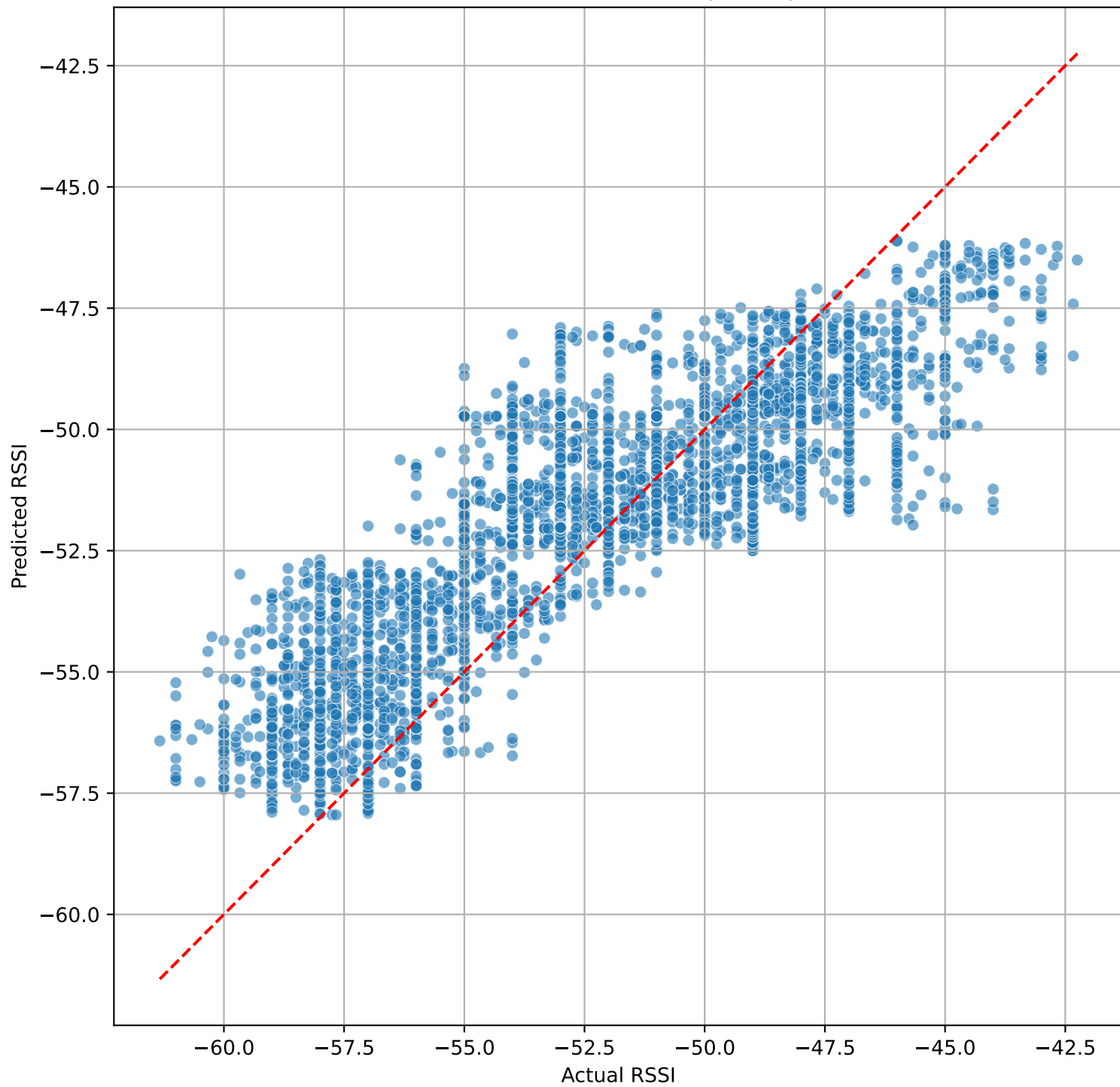
Actual vs Predicted RSSI (Multi-Step (step 36 $\approx$ 3.0 s))



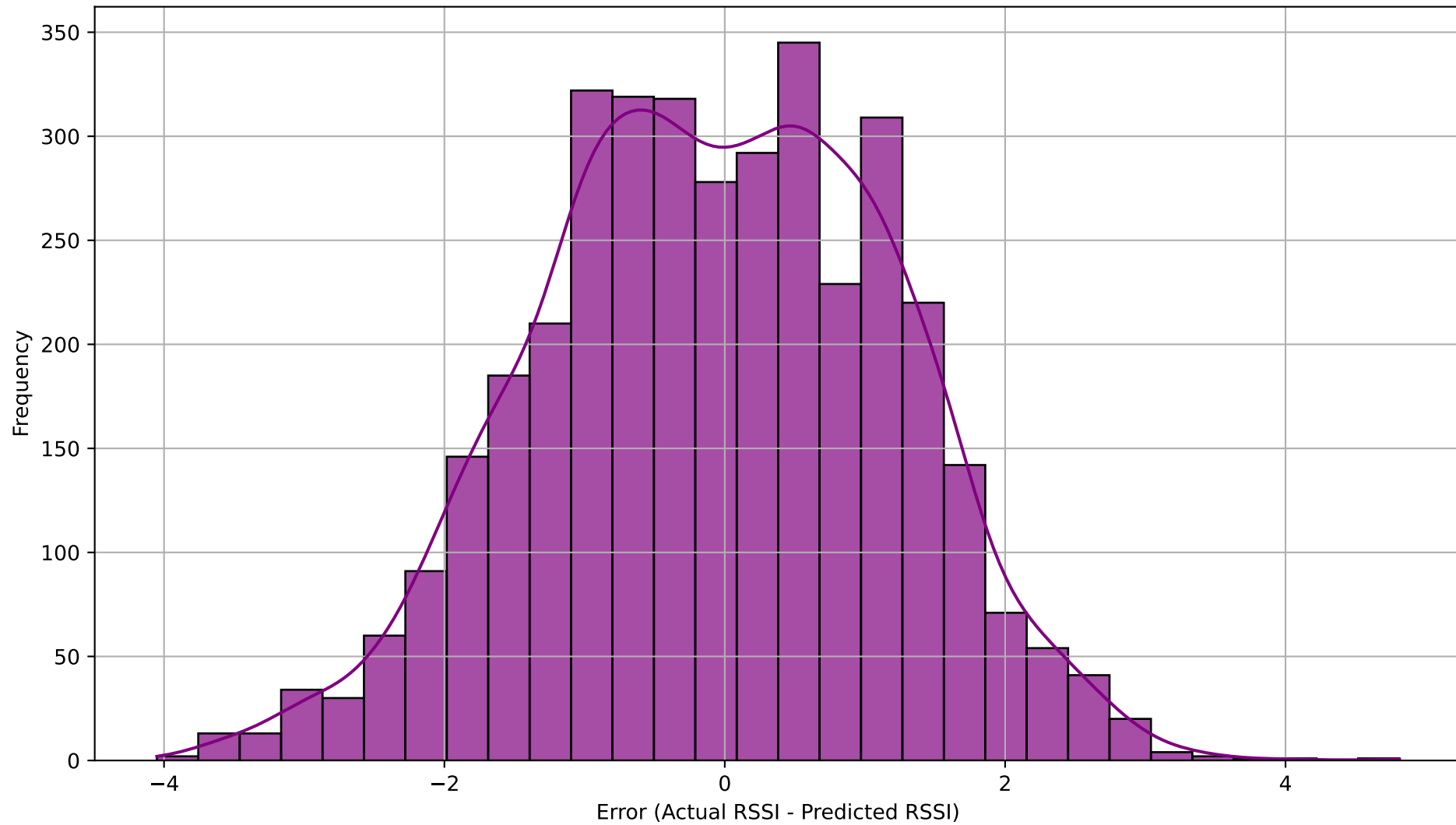
Actual vs Predicted RSSI (First Step)



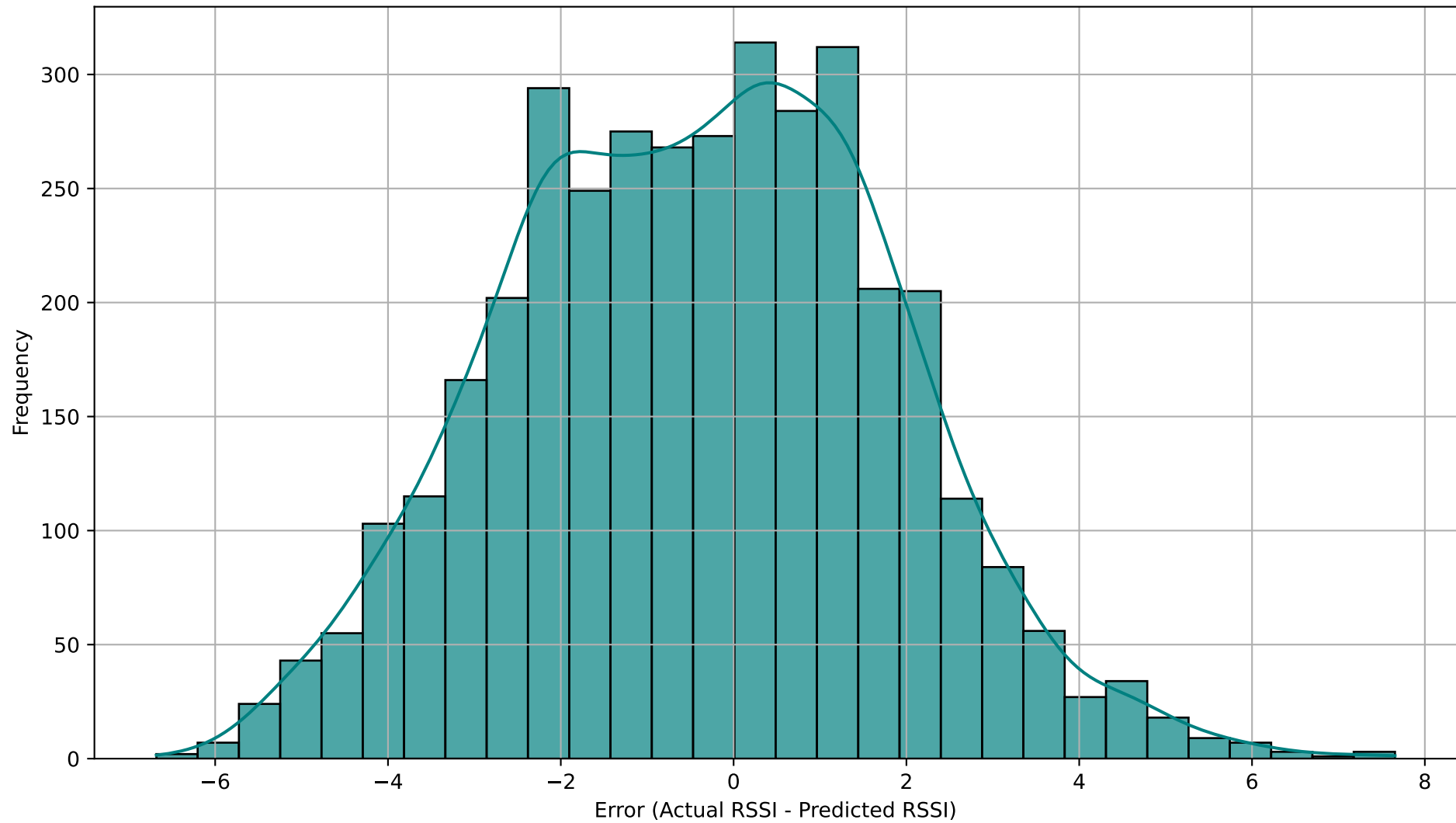
Actual vs Predicted RSSI (Last Step - step 36 $\approx$ 3.0 s)



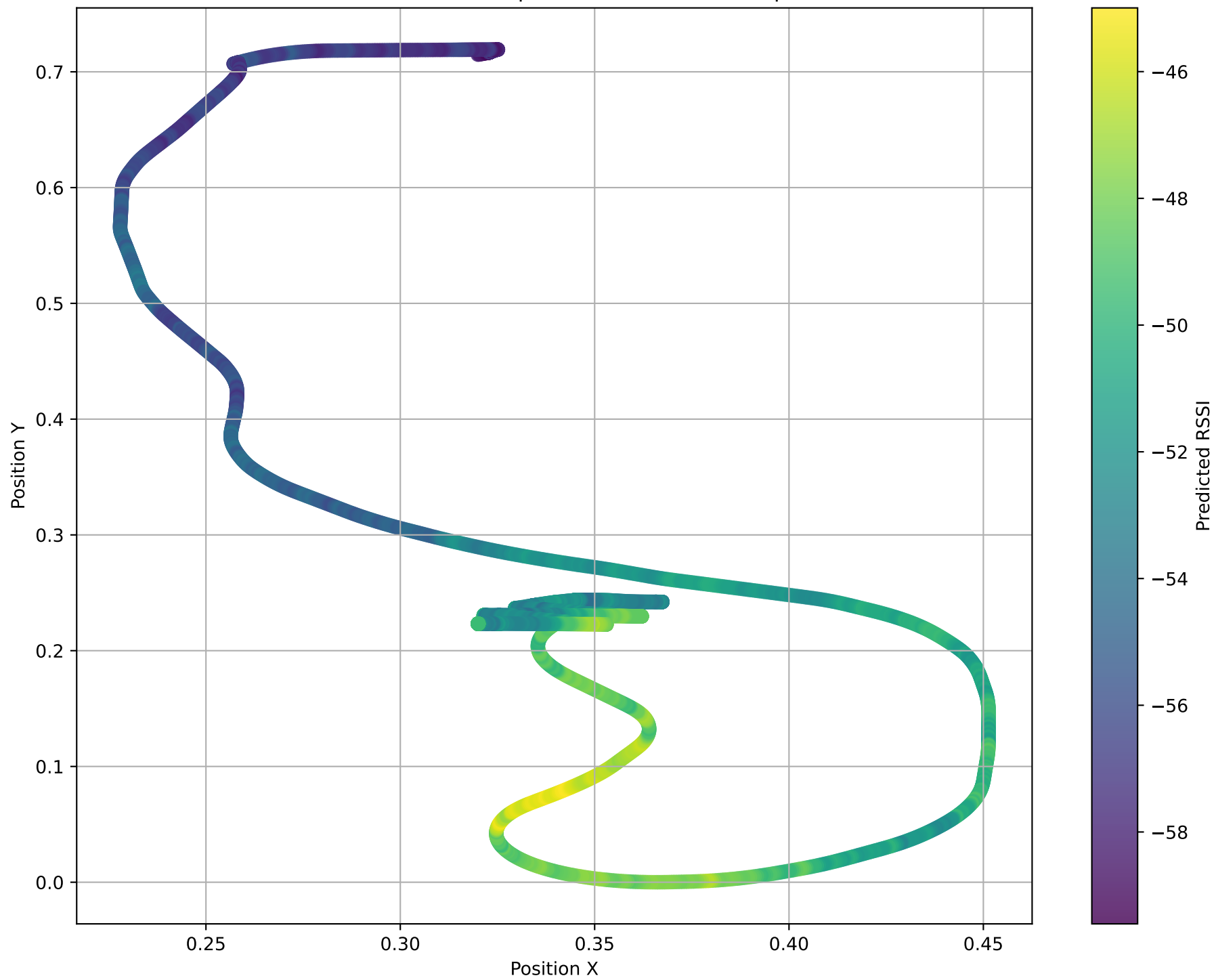
Distribution of Prediction Errors (First Step)



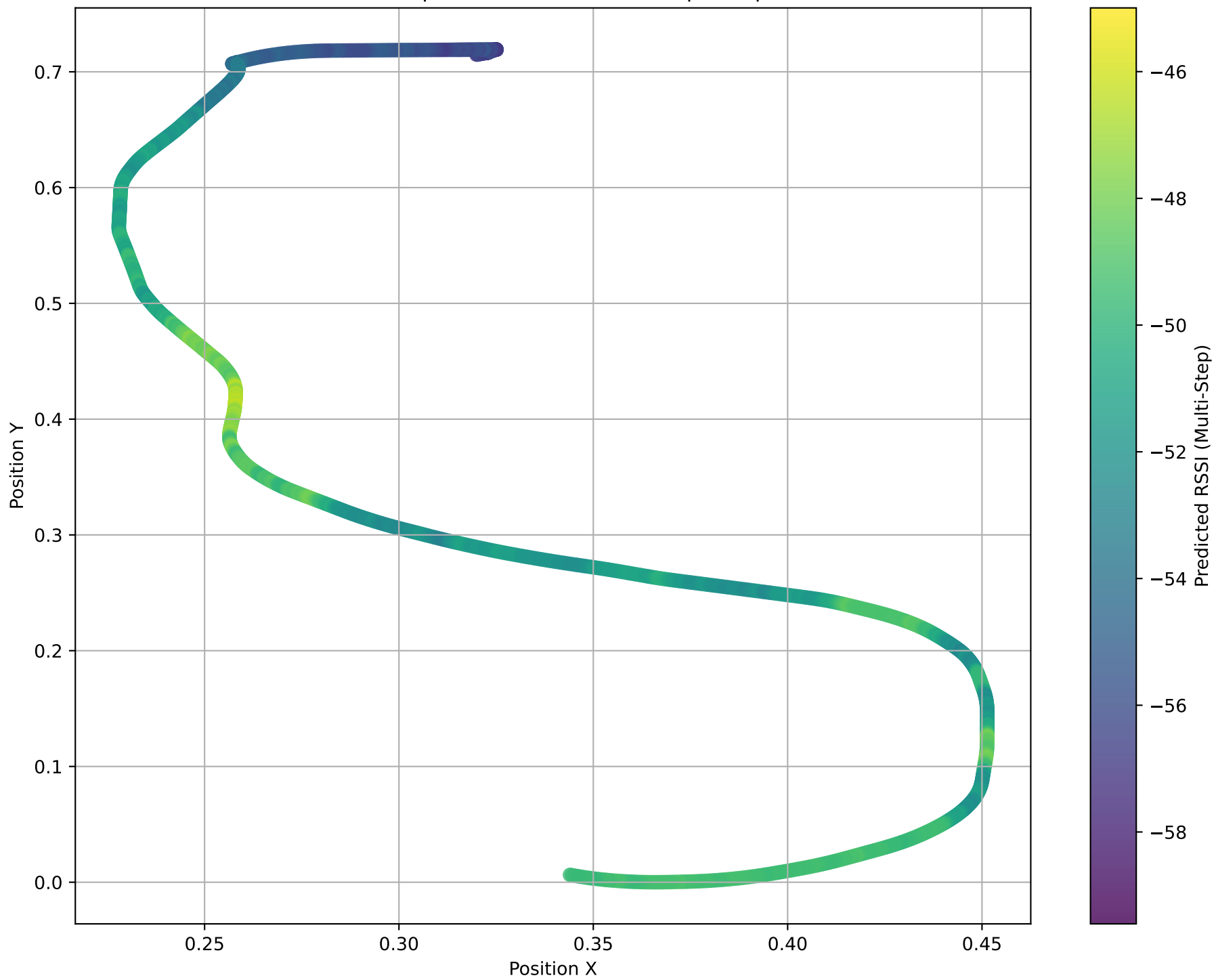
Distribution of Prediction Errors (Last Step - step 36 $\approx$ 3.0 s)



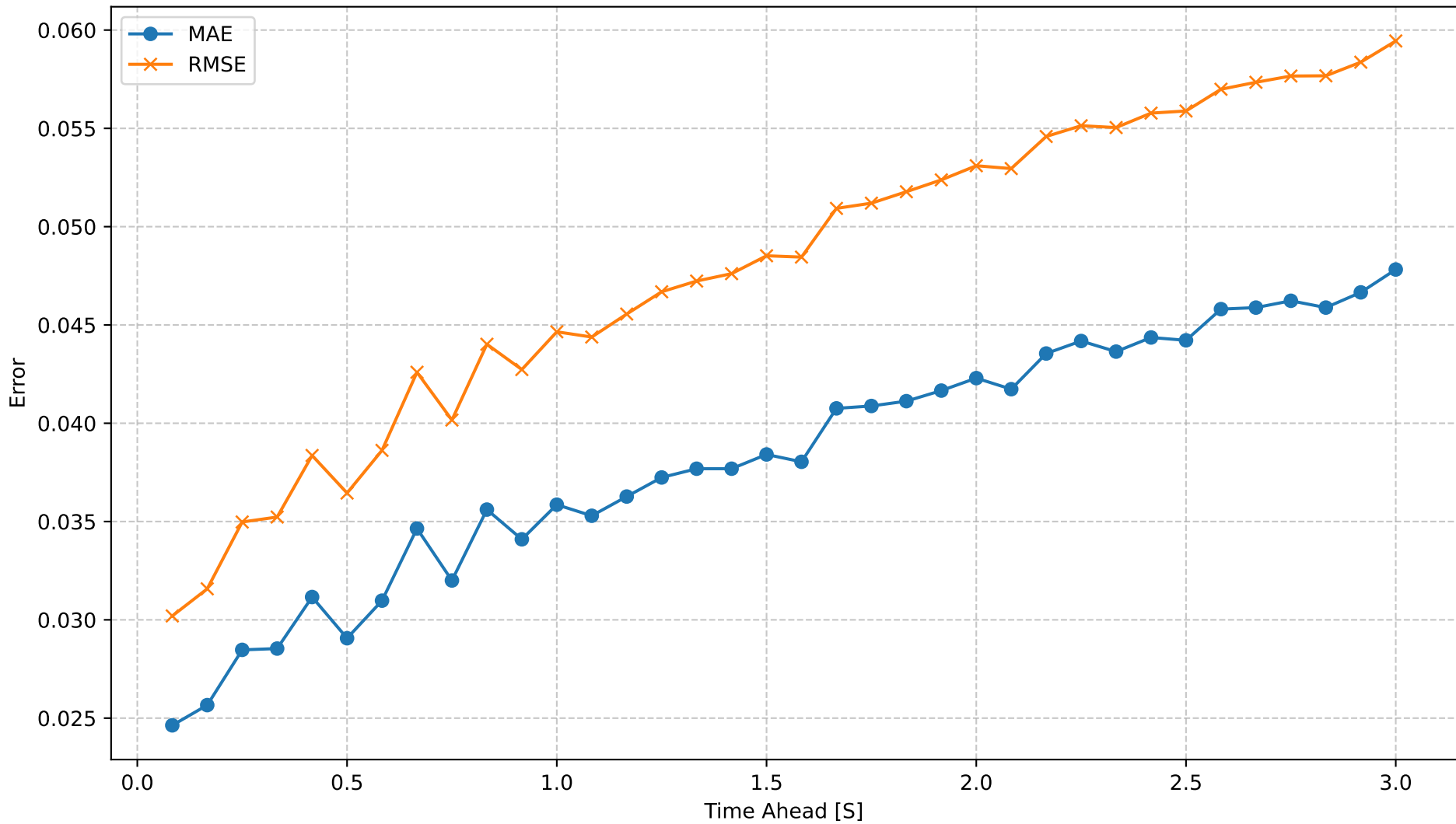
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 36 $\approx$ 3.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet
Modelo: GRU
Forecast steps: 36

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.03
- Mediana : 0.04 / 0.05
- 36º passo : 0.05 / 0.06

Crescimento do erro: 94.1% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 36 (0.05), 35 (0.05), 33 (0.05)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.